

A Unified Inertial Representation for Human and Humanoid Motion

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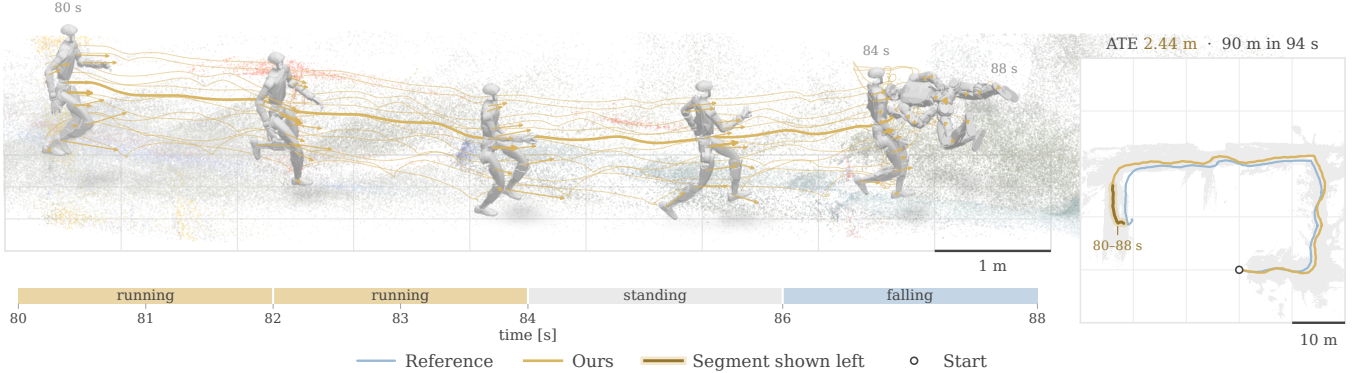


Fig. 1. A single inertial representation learned from a reduced multi-IMU set shared between humans and humanoids. Trained only on real human motion and IK-retargeted G1 trajectories, the model runs zero-shot on a physical Unitree G1: from the robot’s inertial measurements it estimates the pelvis trajectory (right) and retrieves language descriptions of the motion over time (bottom), without dynamics simulation, real-robot training data, or fine-tuning.

Abstract—Inertial measurement units (IMUs) are shared by wearable human motion capture and humanoid robots, but the two embodiments differ in morphology, sensor placement, and sampling rate, and kinematically generated humanoid trajectories lack the contact dynamics, vibration, and sensor imperfections of hardware. We present a single inertial representation that bridges both gaps. A shared inertial interface maps human and humanoid measurements onto twelve anatomically corresponding sites with aligned coordinate conventions, and a continuous-time encoder tokenizes irregularly sampled signals so that one model handles different sensor layouts and rates. A Transformer alternating within-site and across-site attention is trained jointly on real human motion and inverse-kinematics-retargeted G1 trajectories with velocity, trajectory, denoising, and language objectives under hardware-informed sensor randomization. Deployed on a physical Unitree G1 without dynamics simulation, real-robot training trajectories, or fine-tuning, a single checkpoint achieves the lowest absolute velocity error among the evaluated methods on three human datasets, retargeted G1 motion, and real-G1 recordings, and transfers language-grounded motion semantics from humans to the robot, including categories absent from retargeted-G1 training.

I. INTRODUCTION

Inertial measurement units (IMUs) provide a natural sensing interface between humans and humanoid robots. Wearable systems use body-mounted IMUs to capture human motion, while humanoids use the same physical measurements for state estimation and control. The available data, however, are highly asymmetric: human motion datasets are large and behaviorally diverse, whereas collecting comparably rich, accurately labeled data on physical humanoids is expensive. A representation that transfers across embodiments could use human data to supply motion diversity and semantic supervision that are difficult to obtain from robots alone.

Current learned inertial models do not realize this opportunity. Inertial odometry methods typically specialize to one platform or dataset [1], [2], [3], [4], and inertial representation models primarily study transfer across human datasets or tasks [5], [6], [7]. Transfer to a humanoid introduces additional discrepancies in morphology, sensor placement, coordinate convention, and sampling rate. More importantly, kinematically generated trajectories omit the contact dynamics, structural vibration, impact transients, and sensor imperfections present on hardware. The open question is thus whether a single inertial representation can bridge both the embodiment gap from human to humanoid and the sensing gap from kinematic trajectories to a physical robot.

We address this question through zero-shot deployment on a Unitree G1. We train jointly on real human motion and G1 trajectories produced by inverse-kinematics retargeting of human motion. The retargeted data contain kinematics only: we use neither rigid-body dynamics simulation nor real-G1 motion sequences or labels during training. To address the unmodeled hardware effects, we perturb the training signals using randomized embodiment and timing parameters together with hardware-informed models of sensor noise, bias drift, structural vibration, and foot impacts. We deploy the model on hardware without fine-tuning. Hardware evaluation scores also inform checkpoint selection, as detailed in Sec. IV-A. A single checkpoint obtains the lowest absolute velocity error among the evaluated methods on three human datasets, retargeted G1 motion, and real-G1 evaluation recordings. It also transfers language-grounded motion semantics learned from humans to G1 motions, including categories excluded from retargeted-G1 training.

Two design elements make this transfer possible. First, we define a shared inertial interface using twelve anatomically

corresponding sites with aligned coordinate conventions. A continuous-time encoder converts asynchronous, irregularly sampled measurements into fixed-duration tokens, allowing one model to process datasets with different sensor layouts and sampling rates. Second, we learn a task-general representation with a Transformer that alternates within-site attention for local inertial dynamics and across-site attention for whole-body coordination. Joint supervision for velocity, trajectory consistency, denoising, and language alignment enables the same representation to support inertial odometry and language-grounded motion understanding.

Our contributions are:

- We demonstrate zero-shot human-to-humanoid and kinematic-to-real transfer on a physical Unitree G1. Without rigid-body dynamics simulation, real-G1 training trajectories, or hardware fine-tuning, one model achieves the lowest absolute velocity error among the evaluated methods across five datasets and transfers human-supervised motion semantics to the robot.
- We introduce a shared inertial interface combining anatomical and coordinate alignment with continuous-time tokenization for one model across embodiments, sensor configurations, and sampling rates.
- We learn a task-general inertial representation through complementary metric, denoising, and language objectives, supporting inertial odometry and language-grounded motion understanding in one model.

II. RELATED WORK

A. Learned Inertial Odometry

Learned inertial odometry replaces fixed motion assumptions with priors estimated from data. IONet [1] regresses displacement over inertial windows, and RoNIN [2] predicts body-frame velocity for pedestrian navigation. TLIO [3] incorporates learned displacement and uncertainty into a filtering framework. AirIMU [8] learns IMU corrections and their uncertainty, while AirIO [4] improves velocity observability for highly dynamic aerial motion by preserving body-frame measurements. These methods are generally optimized for a particular motion domain. TartanIMU [7] broadens this setting through a shared backbone trained across human and robotic platforms, with platform-specific adaptation. Our work instead defines anatomically corresponding inputs for articulated human and humanoid bodies and learns geometric and semantic motion within one representation.

B. General-Purpose and Language-Aligned IMU Models

Beyond task-specific activity recognition, inertial pre-training seeks representations that transfer across datasets and downstream tasks. LIMU-BERT [9] learns contextual features from unlabeled IMU sequences, and PRIMUS [6] combines self-supervised and multimodal objectives to improve transfer of wearable IMU encoders. Language alignment provides a complementary route to open-vocabulary motion understanding. IMU2CLIP [5] projects human IMU signals into a joint video-text embedding space, while ImageBind [10] includes IMU among six modalities in a shared

representation. More recent models expand this direction: SensorLM [11] learns sensor-language representations from large-scale wearable data, and AURA-MFM [12] aligns IMU with third-person video, motion capture, and text for retrieval and activity recognition. These models advance transferable or language-aligned human sensing, but do not study whether the same representation can provide metric motion estimates and transfer semantics to a humanoid embodiment.

C. Human-to-Humanoid Motion Transfer

Human motion offers a scalable source of behavioral diversity for humanoid learning, but must be reconciled with differences in morphology and physical capability. Earlier work learns a shared latent space for retargeting motion across human and humanoid kinematic structures [13]. Recent systems including H2O [14], OmniH2O [15], and HumanPlus [16] retarget human motion and train control policies in physics simulation for deployment on real humanoids. Human2Humanoid [17] further incorporates morphology-invariant and physics-aware constraints into cross-morphology retargeting. These methods transfer reference motions or control policies to produce robot behavior. We study a distinct transfer problem at the sensing level: learning a common inertial representation from real human measurements and IK-retargeted humanoid kinematics, then applying it to real humanoid measurements without rigid-body dynamics simulation or real-robot training trajectories.

III. METHOD

A. Problem Formulation

Let $\mathcal{S}$ denote a set of anatomically corresponding inertial sites shared by humans and humanoids. For a motion window, site $i \in \mathcal{S}$ provides a timestamped sequence

$$\mathcal{X}_i = \{(t_{i,j}, \mathbf{a}_{i,j}, \boldsymbol{\omega}_{i,j})\}_{j=1}^{T_i}, \quad (1)$$

where $\mathbf{a}_{i,j}, \boldsymbol{\omega}_{i,j} \in \mathbb{R}^3$ are measured specific force (including the gravity contribution) and angular velocity in the local sensor frame. The sequence length T_i may vary with the sensor rate; we retain timestamps instead of imposing a common sampling frequency. The model takes $\mathcal{X} = \{\mathcal{X}_i\}_{i \in \mathcal{S}}$ and predicts body-frame linear velocity $\hat{\mathbf{v}}_{i,j}$ at each site. Pelvis velocity provides the root odometry estimate. In parallel, it maps the window to a feature $\hat{\mathbf{z}}$ aligned with a frozen text embedding space and reconstructs gravity-free kinematic acceleration and clean angular velocity as an auxiliary task. The input and output spaces are identical for both embodiments, so human and robot motion knowledge is shared within one model.

B. Unified Multi-IMU Representation

Humans and humanoids differ in morphology, sensor placement, and sampling rate. We establish correspondence through a common site set and mounting convention, and represent time explicitly. We define a shared set of 12 sites: the pelvis, head, and bilateral upper arms, forearms, upper legs, lower legs, and feet. Streams outside this set are discarded. Human signals come from wearable IMUs or

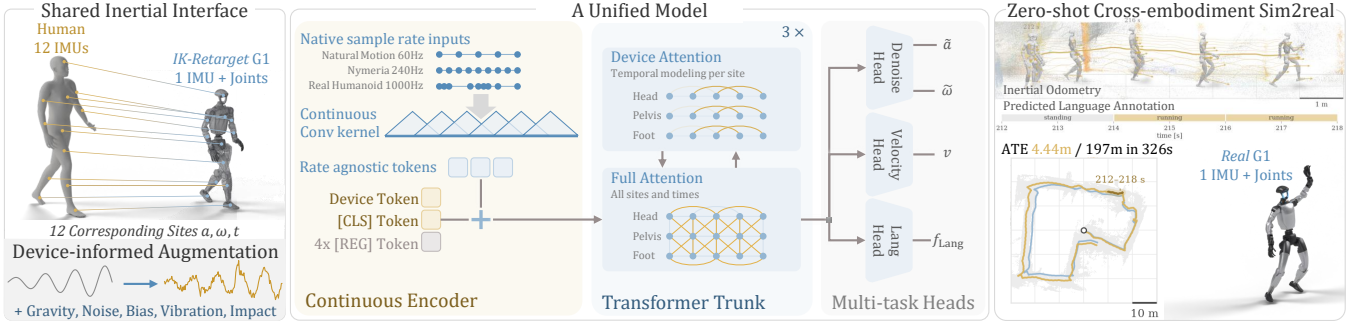


Fig. 2. Overview of the proposed model. Left: a shared inertial interface maps wearable human IMUs and humanoid signals synthesized from one pelvis IMU plus joint states onto twelve anatomically corresponding sites, and device-informed augmentation adds gravity, noise, bias drift, vibration, and impacts during training. Center: a continuous encoder pools measurements at their native rates into rate-agnostic tokens, a Transformer trunk alternates within-site and full attention over all sites and times, and multi-task heads predict denoised inertial signals, body-frame velocity, and a language embedding. Right: the model trained on human motion and IK-retargeted G1 motion is deployed zero-shot on a real G1 for inertial odometry and language-grounded motion annotation.

virtual sensors differentiated from segment kinematics. Each stream retains its anatomical identity, making correspondence explicit without requiring identical hardware.

1) *Humanoid IMU synthesis*: On G1, we transport the pelvis IMU measurements to the selected links using joint positions and velocities. Let $\mathbf{R}_{bi}$ map link- i vectors into pelvis coordinates, and let $\mathbf{r}_i$, $\dot{\mathbf{r}}_i$, and $\ddot{\mathbf{r}}_i$ denote the link origin’s position and its derivatives relative to the pelvis. Placing the base IMU at the pelvis origin gives

$$\begin{aligned}\boldsymbol{\omega}_i &= \mathbf{R}_{bi}^T(\boldsymbol{\omega}_b + \mathbf{J}_{\omega,i}\dot{\mathbf{q}}), \\ \mathbf{a}_i &= \mathbf{R}_{bi}^T(\mathbf{a}_b + \dot{\boldsymbol{\omega}}_b \times \mathbf{r}_i + \boldsymbol{\omega}_b \times (\boldsymbol{\omega}_b \times \mathbf{r}_i) \\ &\quad + 2\boldsymbol{\omega}_b \times \dot{\mathbf{r}}_i + \ddot{\mathbf{r}}_i).\end{aligned}\quad (2)$$

Here $\mathbf{q}$ denotes joint angles and $\mathbf{J}_{\omega,i}$ is the link’s angular Jacobian in pelvis coordinates. Forward kinematics and Jacobians provide the relative link motion; numerical differentiation of the base angular velocity and joint velocities supplies angular and joint accelerations. The transport includes tangential, centripetal, Coriolis, and relative-link acceleration terms, and requires no world pose or root velocity.

For retargeted G1 motion, the base signals are derived from root kinematics; on hardware, they are read from the pelvis IMU. The synthesized accelerations inherit the gravity content of these base signals: gravity-free kinematic inputs yield gravity-free link accelerations, whereas measured specific force retains the gravity contribution at each link.

2) *Coordinate alignment*: Human XSens segment frames define the canonical mounting convention. For each native corpus and site, we estimate $\mathbf{M}_i \in \text{SO}(3)$, which maps canonical vector coordinates into native coordinates. Given the native site-to-world rotation $\mathbf{R}_{wi}$, we use

$$\tilde{\mathbf{R}}_{wi} = \mathbf{R}_{wi}\mathbf{M}_i, \quad \tilde{\mathbf{x}}_i = \mathbf{M}_i^T \mathbf{x}_i, \quad \mathbf{x}_i \in \{\mathbf{a}_i, \boldsymbol{\omega}_i, \mathbf{v}_i\}. \quad (3)$$

To estimate these rotations, we construct anatomical frames from joint positions and compute the chordal mean $\mathbf{S}_{d,i}$ of sensor orientations expressed in anatomical coordinates, $\mathbf{R}_{\text{wanat},i}^T \mathbf{R}_{wi}$ over motion in corpus d . The mounting correction is $\mathbf{M}_i = \mathbf{S}_{d,i}^T \mathbf{S}_{\text{XSens},i}$. It is fixed across recordings, preserves world positions, and rotates the velocity targets

along with the inertial channels. The same G1 correction is used for retargeted and real motion.

3) *Continuous encoding and multi-layer readout*: Each window is packed as per-site streams with explicit timestamps and site boundaries. A shared continuous-kernel encoder uses four triangular basis taps to pool measurements according to their relative times, first over 31.25-ms intervals and then into one 768-dimensional token per occupied 125-ms interval. We add learned site embeddings, four register tokens, and one global [CLS] token per window.

The six Transformer blocks use eight attention heads and timestamp-based rotary positional embeddings. Blocks 1, 3, and 5 attend within each site; blocks 2, 4, and 6 attend over all sites and times in the window. Attention is noncausal. We concatenate the outputs of blocks 2, 4, and 6, independently layer-normalize their 768-dimensional features, and feed the resulting 2,304-dimensional representation to every decoder. Continuous decoders reconstruct gravity-free acceleration, clean angular velocity, and body-frame velocity at the input timestamps. An MLP reads the concatenated [CLS] features to produce the 768-dimensional language embedding. This factorization supports different sampling rates while retaining both local dynamics and whole-body context.

C. Multi-Task Joint Training

We train a single model jointly on human motion and kinematically retargeted G1 motion. The shared trunk is optimized through four complementary objectives:

$$\mathcal{L} = \lambda_{\text{vel}}\mathcal{L}_{\text{vel}} + \lambda_{\text{traj}}\mathcal{L}_{\text{traj}} + \lambda_{\text{lang}}\mathcal{L}_{\text{lang}} + \lambda_{\text{den}}\mathcal{L}_{\text{den}}. \quad (4)$$

1) *Velocity and trajectory consistency*: The velocity loss directly supervises every site’s body-frame prediction, $\mathcal{L}_{\text{vel}} = \mathbb{E}\|\hat{\mathbf{v}} - \mathbf{v}\|_2$. For trajectory supervision, we preintegrate the predicted clean gyroscope with fourth-order Runge–Kutta in single precision, initialized with each site’s ground-truth initial attitude. We stop gradients through this attitude $\hat{\mathbf{R}}$ and integrate $\hat{\mathbf{R}}\hat{\mathbf{v}}$ with the trapezoidal rule to obtain $\hat{\mathbf{p}}$.

For sample pairs (u, v) spanning horizon h , define

$$\begin{aligned}\hat{\mathbf{d}}_{uv} &= \hat{\mathbf{R}}_u^\top (\hat{\mathbf{p}}_v - \hat{\mathbf{p}}_u), \\ \mathbf{d}_{uv} &= \mathbf{R}_u^\top (\mathbf{p}_v - \mathbf{p}_u), \\ \mathcal{L}_{\text{traj}} &= \frac{1}{|\mathcal{H}|} \sum_{h \in \mathcal{H}} \mathbb{E}_{(u,v) \in \mathcal{W}_h} \left\| \frac{\hat{\mathbf{d}}_{uv} - \mathbf{d}_{uv}}{t_v - t_u} \right\|_2^2,\end{aligned}\quad (5)$$

where $\mathcal{H}$ contains the available horizons from $\{0.25, 1, 2\}$ s and $\mathcal{W}_h$ contains non-overlapping intervals on each site’s timestamp grid. This loss penalizes relative displacement drift per unit time and trains the velocity head and shared trunk; it does not backpropagate through the gyroscope head. Rotation errors are logged as metrics rather than optimized by an additional trajectory loss.

2) *Language alignment and denoising*: A frozen CLIP ViT-L/14@336px text encoder [18] maps motion descriptions to 768-dimensional targets $\mathbf{z}$. The global inertial feature is regressed with

$$\mathcal{L}_{\text{lang}} = \mathbb{E}_{\text{annotated}}[1 - \cos(\hat{\mathbf{z}}, \mathbf{z})]. \quad (6)$$

Only windows with a valid language target contribute. Natural Motion has no language annotations; Nymeria windows contribute only when their full extent is covered by annotation spans sharing one target embedding. Other windows still train the motion objectives. The denoising loss sums the mean Euclidean errors of reconstructed gravity-free acceleration and clean angular velocity against their augmented clean targets. It therefore supervises gravity removal as well as rejection of sensor disturbances. Training uses corrupted, unmasked inputs for all four objectives.

3) *Adaptive loss weights*: We fix $\lambda_{\text{den}} = \lambda_{\text{lang}} = 1$. For velocity and trajectory, we measure gradient norms at a shared feature probe and compute the ratio of the denoising gradient norm to each task’s gradient norm (with 10^{-6} added to the denominator). Each task weight is an exponential moving average of this ratio with decay 0.99, clipped after every update to $[0.02, 50]$. The first valid ratio initializes the weight; it is held when that task has no valid gradient.

D. Hardware-Informed Sim-to-Real Augmentation

Kinematic retargeting captures motion but omits the sensor noise and mechanical disturbances present on hardware. Our augmentation pipeline accounts for these sensing differences while varying body scale and motion timing. We first convert gravity-free training accelerations to specific force by adding the gravity contribution in each site’s local frame. We then randomize embodiment scale and motion speed. A length factor k scales positions, linear velocities, and kinematic accelerations. A speed factor s compresses relative timestamps by $1/s$, scales velocities and angular rates by s , and scales kinematic accelerations by s^2 . Clean targets and velocity labels follow these changes; gravity is unchanged.

We next add white sensor noise, bias drift, structural vibration, and sparse impact transients to both human and retargeted-G1 inputs. The latter two use disturbance statistics from G1 hardware. Vibration is colored noise shaped by Lorentzian resonance modes and modulated by a smoothed

TABLE I
TRAINING RANDOMIZATION SETTINGS. VIBRATION AND IMPACT AMPLITUDES ARE BASE DRAWS.

Randomization	Setting
Gravity magnitude	$\mathcal{U}(9.76, 9.83)$ m/s ²
Human / G1 scale	$\mathcal{U}(0.75, 1.0)$ / $\mathcal{U}(1.0, 1.33)$
Human / G1 speed	$\mathcal{U}(0.7, 1.3)$ / $\mathcal{U}(0.9, 1.6)$
Human rate target	$\mathcal{U}(120, 240)$ Hz, integer decimation
Timestamp jitter	Requested spacing CV $\mathcal{U}(0, 0.15)$; bounded to preserve order
Acc. noise density	10^{-2} (m/s ²)/ $\sqrt{\text{Hz}}$
Gyro noise density	3×10^{-4} (rad/s)/ $\sqrt{\text{Hz}}$
Acc. bias walk	5×10^{-4} (m/s ²)/ $\sqrt{s}$
Gyro bias walk	5×10^{-5} (rad/s)/ $\sqrt{s}$
Vibration	1–3 modes; centers 5–60 Hz; bandwidths 2–15 Hz; 3-Hz high-pass; base RMS 0.3–3.0 m/s ²
Impact	Rate 0.5–4.0 events/s; base peak 3–30 m/s ² ; decay 15–50 ms; ring 8–30 Hz
Gyro / acc. ratio	Vibration 0.015–0.05; impact 0.02–0.08 (rad/s)/(m/s ²)

clean-gyroscope motion envelope. Impacts use Poisson event counts, motion-weighted event times, and exponentially decaying sinusoidal ring-downs. Amplitude scales are sampled log-uniformly; per-site gains and motion modulation change the realized amplitudes, and impact peaks additionally receive log-normal multipliers. Table I lists the settings.

Human rate reduction draws a target f_* uniformly from 120–240 Hz. Given the time-warped rate f , we round f/f_* to an integer factor $m \geq 1$ and decimate to f/m Hz. The realized rate therefore need not equal the target. Anti-alias filtering applies to both measured and clean inertial channels before subsampling; pose and velocity labels are subsampled without filtering. Retargeted G1 uses no decimation; its 120-Hz source clock becomes approximately 108–192 Hz through time warping. Finally, both embodiments receive bounded timestamp jitter, with amplitudes unchanged, and are pruned to the shared site set.

Additive disturbances affect only the inputs; denoising targets remain clean versions of the scaled, time-warped, and, where applicable, filtered signals. Human and retargeted-G1 validation data retain gravity, noise, bias, vibration, and impacts but omit scale, speed, decimation, and timestamp randomization. Real-G1 evaluation inputs are only pruned to the shared sites and receive no synthetic corruption.

We use hardware-informed zero-shot transfer to mean deployment without real-G1 gradient training or fine-tuning. Hardware recordings inform the corruption family, and real-G1 evaluation scores also inform checkpoint selection, as detailed in Sec. IV-A. The reported hardware results therefore include model selection on that evaluation set.

IV. EXPERIMENTS

We evaluate one jointly trained model on human and humanoid odometry and IMU–language retrieval, then ablate

TABLE II
DATASETS FOR FOUNDATIONIMU TRAINING AND EVALUATION.

Dataset	Emb.	Motion	Real IMU	Freq.	Hours	Lang.	Train
Nymeria	Human	Real		240	300+	✓	✓
Natural Motion	Human	Real		240	40.6		✓
BONES-Seed	Human	Real		120	144	✓	✓
BONES-Seed	G1	Retarget		120	144	✓	✓
Real-Trajectory	G1	Real	✓	200	0.7		

its training data, input representation, and architecture.

A. Experimental Setup

1) *Datasets*: Training combines Nymeria [19], Natural Motion [20], and BONES-Seed [21] human motion with kinematically retargeted BONES-Seed (G1) motion (Table II). Categories withheld from G1 training but present in human training test semantic transfer. Real-G1 recordings (Real-Trajectory) inform disturbance modeling and checkpoint selection but are excluded from gradient training.

2) *Baselines*: For odometry, we train pelvis-IMU baselines TLIO [3], AirIMU [8], and AirIO [4] separately on each dataset, plus an AirIO variant with 11 relative joint positions and velocities. We evaluate the released TartanIMU [7] human head without retraining on sequences exceeding 10 s.

For retrieval, IMU2CLIP [5] is trained per dataset with the pelvis IMU; ImageBind [10] uses its released checkpoint, head IMU, and text encoder. Our model and IMU2CLIP use 4.096-s windows. ImageBind uses 10-s windows resampled to 2,000 samples, giving different temporal context and annotation coverage.

3) *Metrics*: Odometry metrics pool pelvis samples across recordings. **Absolute velocity error** (AVE, m/s) is

$$\text{AVE} = \frac{1}{T} \sum_{t=1}^T \|\hat{\mathbf{v}}_t - \mathbf{v}_t\|_2. \quad (7)$$

Absolute trajectory error (ATE, m) is the position RMSE after integrating velocity from the ground-truth initial pose:

$$\hat{\mathbf{p}}_t = \mathbf{p}_0 + \int_0^t \mathbf{R}_\tau \hat{\mathbf{v}}_\tau d\tau, \quad \text{ATE} = \sqrt{\frac{1}{T} \sum_{t=1}^T \|\hat{\mathbf{p}}_t - \mathbf{p}_t\|^2}. \quad (8)$$

Here $\mathbf{R}_t$ is ground-truth attitude for ATE_{GT} and gyro-integrated attitude for ATE_{Pred} , with no filter updates or trajectory alignment. Our model integrates the measured gyroscope; both AirIO variants use AirIMU-corrected gyroscopes. TLIO aligns its inputs with ground-truth attitude. Pelvis-only AirIO takes ground-truth attitude for AVE/ ATE_{GT} and integrated attitude for ATE_{Pred} ; its joint-input variant takes no attitude.

Retrieval ranks cosine similarities within each dataset. **Text retrieval** reports recall at 1 (R@1) against 19 random distractors, averaged over eight draws. **Motion retrieval** reports average precision over all annotated windows, averaged over distinct text queries (mAP). Collapse assigns all windows the normalized mean of the evaluation text targets.

4) *Implementation Details*: We train for 64 epochs on nominal 4.096-s windows with bfloat16 precision and 128 windows per GPU on eight GPUs (global batch 1024). Muon updates matrix parameters at 2×10^{-3} ; AdamW updates the rest at 5×10^{-3} . Both use weight decay 10^{-3} and gradient clipping at norm 1. Architecture, losses, and augmentations are specified in Sec. III.

All Ours rows use exponential-moving-average (EMA) weights after 20 epochs (step 6420), selected by screening all 64 checkpoints on real G1 and checking the seen-category retargeted-G1 split. Hardware scores thus include evaluation-set selection, with no real-G1 gradient training or fine-tuning.

B. Inertial Odometry

Our model has the lowest AVE and ATE_{GT} on all five datasets (Table III), reaching 0.024 m/s AVE on retargeted G1 and 0.099 m/s on real G1. It also leads ATE_{Pred} on four datasets; TLIO is slightly better on Nymeria (17.76 versus 18.20 m). The larger errors with integrated attitude show the remaining orientation drift. Fig. 3 shows representative trajectories.

C. IMU–Language Retrieval

Our model leads both metrics on BONES-Seed human motion and both G1 splits (Table IV). On categories excluded from G1 training, it achieves 53.8% R@1 and 11.3% mAP, versus 17.3% and 0.5% for IMU2CLIP, supporting semantic transfer from human motion. On Nymeria, IMU2CLIP has higher R@1 (16.2% versus 14.9%), while our model has higher mAP. Fig. 4 shows a temporal retrieval example.

D. Ablation Studies

Table V reports ablations on Nymeria, retargeted G1, and real G1. Input and architecture variants use 6420 optimizer updates, matching Ours.

1) *Human-to-Humanoid Transfer*: At 20 epochs, joint training reduces ATE_{GT} relative to humanoid-only training on all three datasets, including 5.90 to 3.15 m on real G1. Both G1 trajectory metrics improve, while Nymeria ATE_{Pred} increases slightly. The joint run receives 6420 updates versus 800 for humanoid-only training, so this comparison combines data diversity with additional optimization. Human and humanoid embeddings overlap in Fig. 5.

2) *Multi-IMU Representation*: We compare the shared 12-site input with pelvis-only input and pelvis IMU plus 11 relative joint positions and velocities. The joint-input variant uses disturbed joint measurements as separate streams in the same trunk. Our G1 multi-IMU input is synthesized from pelvis IMU and joint states (Sec. III-B).

Pelvis-only input raises real-G1 ATE_{GT} from 3.15 to 8.11 m. Direct joints give lower ATE_{GT} on Nymeria and real G1, but higher real-G1 ATE_{Pred} (18.36 versus 11.50 m). Multi-IMU input thus improves on pelvis-only input; its advantage over direct joints depends on the metric.

TABLE III
INERTIAL ODOMETRY (AVE [M/S] ↓, ATE [M] ↓) WITH PELVIS IMU, PELVIS IMU PLUS JOINTS, AND SHARED MULTI-IMU INPUTS (OURS). BOLD MARKS COLUMN MINIMA.

Method	Human									Humanoid (G1)					
	Nymeria			Natural Motion			BONES-Seed (Human)			BONES-Seed (G1)			Real-Trajectory		
	AVE	ATE _{GT}	ATE _{Pred}	AVE	ATE _{GT}	ATE _{Pred}	AVE	ATE _{GT}	ATE _{Pred}	AVE	ATE _{GT}	ATE _{Pred}	AVE	ATE _{GT}	ATE _{Pred}
<i>Pelvis IMU only</i>															
TLIO [†] [3]	0.122	8.149	17.76	0.189	40.72	81.31	0.245	0.717	0.719	0.173	0.761	0.762	0.371	17.86	17.86
AirIMU [†] [8]	12.62	—	—	109.3	—	—	0.212	9.472	505.5	0.281	11.83	306.6	6.231	731.1	—
AirIO [†] [4]	0.069	7.481	113.9	0.280	158.1	157.9	0.140	0.530	1.918	0.118	0.720	1.690	0.350	16.82	20.48
TartanIMU [7]	0.301	37.74	35.53	0.332	155.8	156.3	0.328*	2.970*	2.964*	0.219*	2.176*	2.176*	0.391	18.94	18.96
<i>Pelvis IMU + joint positions and velocities</i>															
AirIO w/ joints [†]	0.060	6.912	39.64	0.080	14.11	209.2	0.205	0.907	1.652	0.123	0.620	1.565	0.190	5.976	22.25
Ours	0.026	2.468	18.20	0.049	7.456	50.54	0.031	0.113	0.148	0.024	0.101	0.127	0.099	3.153	11.50

[†]Trained separately on each dataset. *Only sequences longer than 10 seconds are evaluated.

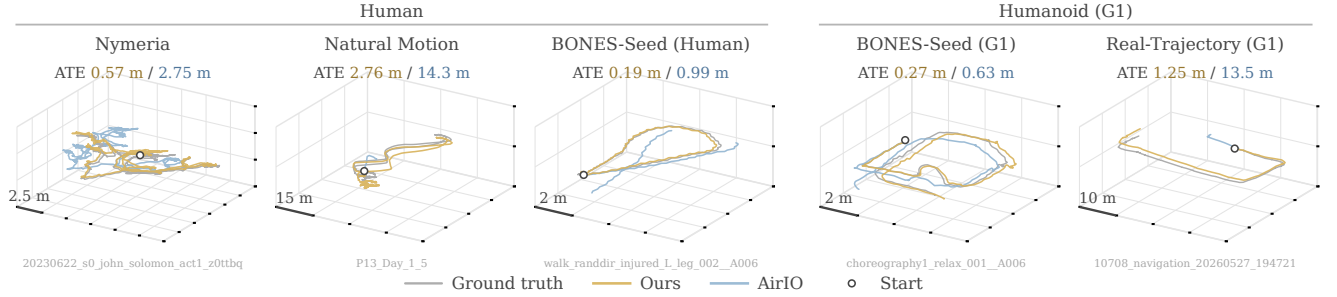


Fig. 3. Odometry on one evaluation recording per dataset, with equal 3D axis scales. Pelvis velocities from our model (gold) and AirIO (blue) are integrated with ground-truth attitude; gray denotes ground truth. Panel values are per-recording ATE_{GT} (Ours/AirIO), distinct from the pooled table scores.

TABLE IV
IMU–LANGUAGE RETRIEVAL (R@1 / MAP, % ↑). BOLD MARKS THE BEST VALUE FOR EACH METRIC AND SPLIT.

Method	Human		Humanoid (G1)	
	Nymeria	BONES-Seed	Seen cat.	Unseen cat.
Collapse	4.4 / 0.1	19.7 / 0.2	21.9 / 0.8	9.8 / 0.1
ImageBind [10]	8.9 / 0.1	7.1 / 0.4	11.4 / 1.6	4.6 / 0.1
IMU2CLIP [†] [5]	16.2 / 0.4	51.8 / 5.2	36.1 / 11.2	17.3 / 0.5
Ours	14.9 / 0.6	60.7 / 15.9	63.4 / 32.9	53.8 / 11.3

[†]Trained separately on each dataset.

3) *Model Design*: Replacing alternating attention with six full-window blocks preserves parameter count. It leaves Nymeria ATE_{GT} nearly unchanged and improves ATE_{Pred} (14.31 versus 18.20 m), but slightly worsens both G1 metrics.

Replacing continuous tokenization with a strided convolution over 25-sample patches raises ATE_{GT} on all three datasets, including 3.15 to 17.37 m on real G1. Fixed-sample patches span different durations at different rates, supporting the use of explicit time in the shared encoder.

E. Compute Allocation and Data Diversity

We vary width and depth across seven models (2.83–274.25 M parameters), halving the learning rates for the largest. Following Chinchilla [22], 235 EMA checkpoints

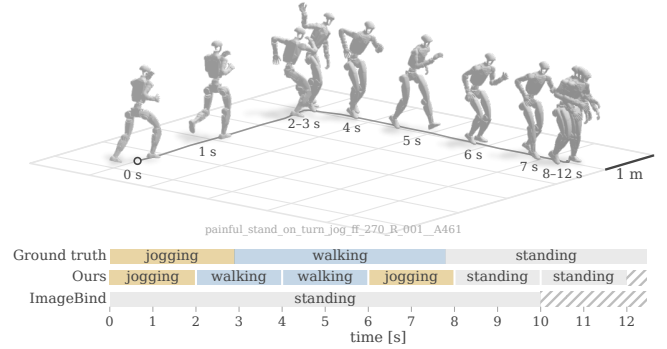


Fig. 4. IMU–language retrieval on a retargeted G1 recording. Top: poses at one-second intervals. Bottom: annotated and retrieved categories. Ours uses 2-s windows here; ImageBind uses 10-s windows. Hatching marks intervals without a complete window.

from nine runs, including three seeds at 14.58 M, fit

$$L(N, D) = E + \frac{A}{N^\alpha} + \frac{B}{D^\beta}. \quad (9)$$

L is shared-site validation velocity error, including real G1; N counts all model parameters. At s updates, $D = s(8 \cdot 128) \cdot 401$ counts nominal tokens from all eight GPUs, including repeated windows. Fitting log loss with a Huber objective gives $A = 192.53$, $B = 1.4094 \times 10^4$, $\alpha = 0.6164$, $\beta = 0.6657$, and $E = 0.02362$ m/s, using individual parameters and tokens.

TABLE V

ODOMETRY ABLATIONS (ATE [M] ↓) OF TRAINING EMBODIMENT, INPUT, DATA SCALE, AND ENCODER DESIGN. TRAINING BUDGETS ARE DETAILED IN SEC. IV-D. BOLD MARKS COLUMN MINIMA.

Variant	Nymeria		BONES-Seed (G1)		Real-Trajectory	
	ATE _{GT}	ATE _{Pred}	ATE _{GT}	ATE _{Pred}	ATE _{GT}	ATE _{Pred}
<i>Training embodiment</i>						
Humanoid-only	8.041	17.50	0.162	0.180	5.901	13.15
<i>Input representation</i>						
Single pelvis IMU	3.592	16.66	0.330	0.335	8.114	12.81
Pelvis IMU + joints	2.283	16.12	0.136	0.164	3.513	18.36
<i>Training-data scale</i>						
1/64 of data	4.882	18.40	0.154	0.166	4.369	12.44
1/16 of data	3.500	16.93	0.111	0.131	3.792	11.95
1/4 of data	2.847	14.64	0.099	0.121	3.716	11.98
<i>Model design</i>						
w/o Alternating Attn.	2.466	14.31	0.110	0.128	3.530	11.71
w/o Continuous Enc.	31.42	31.51	1.189	1.189	17.37	17.67
Ours	2.468	18.20	0.101	0.127	3.153	11.50

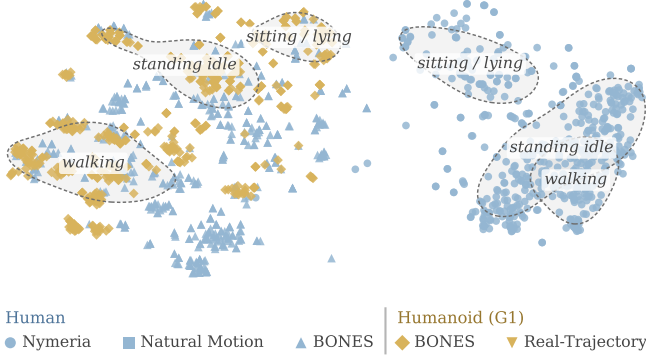


Fig. 5. t-SNE of human (blue) and humanoid (gold) [CLS] embeddings. Dashed regions enclose 60% of windows in each labeled motion category; corresponding BONES human and G1 categories overlap.

Under the compute proxy $C = 6ND$, the fit gives $N_{\text{opt}} \propto C^{0.52}$ and $D_{\text{opt}} \propto C^{0.48}$ (Fig. 6). Each contour trades model size against exposure at fixed compute; for example, 10^{18} FLOP favors 29.0 M parameters and 5.75 B processed tokens. These predictions optimize the fitted validation loss, include extrapolation, and need not optimize hardware transfer. The law has no separate saturation parameter for repeated data.

Separately, Table V varies unique training sequences (1/64, 1/16, 1/4, full) at 55.56 M parameters and within 2.4% of the same update budget. More unique data monotonically reduces ATE_{GT} on Nymeria and real G1; quarter-data training has the lowest retargeted-G1 trajectory errors.

V. CONCLUSIONS

We presented a unified multi-IMU representation for human and humanoid motion. A shared interface of twelve anatomically corresponding sites, aligned mounting conventions, and continuous-time tokenization lets one model consume wearable human measurements and humanoid signals synthesized from a pelvis IMU and joint states at different sampling rates. A Transformer with alternating within-site and across-site attention is trained jointly on real human motion and kinematically retargeted G1 motion with velocity,

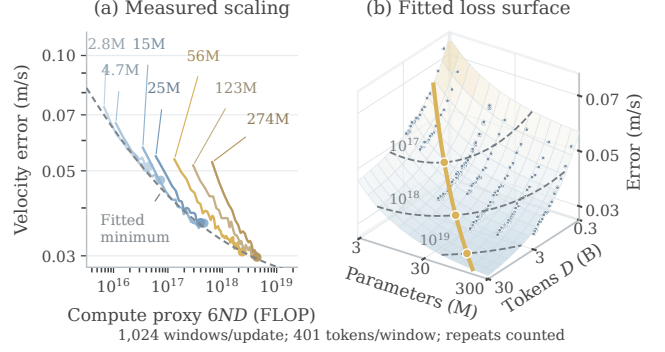


Fig. 6. Compute allocation with a global batch of 1,024 windows and 401 nominal tokens/window, including repeats. Both panels use 235 EMA checkpoints. (a) Seed-averaged measurements; dashed line: fitted minimum. (b) Eq. 9 and individual observations (blue). Gray contours hold $6ND$ at 10^{17} – 10^{19} FLOP; gold marks fitted minima and their path. The surface includes extrapolation.

trajectory-consistency, denoising, and language-alignment objectives under hardware-informed sensor randomization. Without dynamics simulation, real-G1 training trajectories, or fine-tuning, a single checkpoint attains the lowest absolute velocity error among the evaluated methods on five datasets and transfers language-grounded motion semantics from humans to the robot, including categories withheld from G1 training.

Several limitations remain. Hardware evaluation covers a single platform, the Unitree G1, with 0.7 hours of recordings, and checkpoint selection uses real-G1 scores, so the hardware results include model selection on that set. Language supervision is partly synthetic: retargeted-G1 windows inherit the temporal labels of their source human takes, and the vocabulary is narrower than free-form narration. Trajectory error with gyro-integrated attitude also remains well above that with ground-truth attitude, reflecting uncorrected orientation drift. Future work will extend training and evaluation to additional humanoid platforms and fuse the learned velocity with a filter for closed-loop state estimation and control.

REFERENCES

- [1] C. Chen, X. Lu, A. Markham, and N. Trigoni, “IONet: Learning to cure the curse of drift in inertial odometry,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 32, no. 1, 2018, pp. 6468–6476.
- [2] S. Herath, H. Yan, and Y. Furukawa, “RoNIN: Robust neural inertial navigation in the wild: Benchmark, evaluations, & new methods,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2020, pp. 3146–3152.
- [3] W. Liu, D. Caruso, E. Ilg, J. Dong, A. I. Mourikis, K. Daniilidis, V. Kumar, and J. Engel, “TLIO: Tight learned inertial odometry,” *IEEE Robotics and Automation Letters*, vol. 5, no. 4, pp. 5653–5660, 2020.
- [4] Y. Qiu, C. Xu, Y. Chen, S. Zhao, J. Geng, and S. Scherer, “AirIO: Learning inertial odometry with enhanced IMU feature observability,” *IEEE Robotics and Automation Letters*, 2025, arXiv preprint arXiv:2501.15659.
- [5] S. Moon, A. Madotto, Z. Lin, A. Saraf, A. Bearman, and B. Dama-vandi, “IMU2CLIP: Language-grounded motion sensor translation with multimodal contrastive learning,” in *Findings of the Association for Computational Linguistics: EMNLP 2023*, 2023, pp. 13 246–13 253.

- [6] A. M. Das, C. I. Tang, F. Kawsar, and M. Malekzadeh, "PRIMUS: Pretraining IMU encoders with multimodal self-supervision," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2025, arXiv preprint arXiv:2411.15127.
- [7] S. Zhao, S. Zhou, R. Blanchard, Y. Qiu, W. Wang, and S. Scherer, "Tartan IMU: A light foundation model for inertial positioning in robotics," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025, pp. 22 520–22 529.
- [8] Y. Qiu, C. Xu, Y. Chen, S. Zhao, J. Geng, and S. Scherer, "AirIMU: Learning uncertainty propagation for inertial odometry," *Robotics and Autonomous Systems*, 2025, arXiv preprint arXiv:2310.04915.
- [9] H. Xu, P. Zhou, R. Tan, M. Li, and G. Shen, "LIMU-BERT: Unleashing the potential of unlabeled data for IMU sensing applications," in *Proceedings of the 19th ACM Conference on Embedded Networked Sensor Systems (SenSys)*, 2021.
- [10] R. Girdhar, A. El-Nouby, Z. Liu, M. Singh, K. V. Alwala, A. Joulin, and I. Misra, "ImageBind: One embedding space to bind them all," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 15 180–15 190.
- [11] Y. Zhang, K. Ayush, S. Qiao, A. A. Heydari, G. Narayanswamy, M. A. Xu, A. A. Metwally, S. Xu, J. Garrison, X. Xu, T. Althoff, Y. Liu, P. Kohli, J. Zhan, M. Malhotra, S. Patel, C. Mascolo, X. Liu, D. McDuff, and Y. Yang, "SensorLM: Learning the language of wearable sensors," in *Advances in Neural Information Processing Systems*, vol. 38, 2025.
- [12] K. Matsuishi, K. Ukita, and T. Okita, "Multimodal foundation model for cross-modal retrieval and activity recognition tasks," *International Journal of Activity and Behavior Computing*, vol. 2025, no. 3, pp. 1–25, 2025.
- [13] S. Choi, M. Pan, and J. Kim, "Nonparametric motion retargeting for humanoid robots on shared latent space," in *Robotics: Science and Systems*, 2020.
- [14] T. He, Z. Luo, W. Xiao, C. Zhang, K. Kitani, C. Liu, and G. Shi, "Learning human-to-humanoid real-time whole-body teleoperation," *arXiv preprint arXiv:2403.04436*, 2024.
- [15] T. He, Z. Luo, X. He, W. Xiao, C. Zhang, W. Zhang, K. Kitani, C. Liu, and G. Shi, "OmniH2O: Universal and dexterous human-to-humanoid whole-body teleoperation and learning," *arXiv preprint arXiv:2406.08858*, 2024.
- [16] Z. Fu, Q. Zhao, Q. Wu, G. Wetzstein, and C. Finn, "HumanPlus: Humanoid shadowing and imitation from humans," *arXiv preprint arXiv:2406.10454*, 2024.
- [17] T. Huang, F. Yuan, J. Gu, S. Fang, X. Zhang, Y. Wang, W. Gao, and S. Zhang, "Human2Humanoid: Physics-aware cross-morphology motion retargeting for humanoid robots," *arXiv preprint arXiv:2606.03476*, 2026.
- [18] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, "Learning transferable visual models from natural language supervision," in *Proceedings of the 38th International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 139, 2021, pp. 8748–8763.
- [19] L. Ma, Y. Ye, F. Hong, V. Guzov, Y. Jiang, R. Postyeni, L. Pesqueira, A. Gamino, V. Baiyya, H. J. Kim, K. Bailey, D. S. Fosas, C. K. Liu, Z. Liu, J. Engel, R. De Nardi, and R. Newcombe, "Nymeria: A massive collection of multimodal egocentric daily motion in the wild," in *European Conference on Computer Vision (ECCV)*, 2024.
- [20] J. H. Geissinger and A. T. Asbeck, "Motion inference using sparse inertial sensors, self-supervised learning, and a new dataset of unscripted human motion," *Sensors*, vol. 20, no. 21, p. 6330, 2020.
- [21] Bones Studio, "BONES-SEED: Skeletal Everyday Embodied Dataset," <https://huggingface.co>, 2026, accessed: 2026-08-16.
- [22] J. Hoffmann, S. Borgeaud, A. Mensch, E. Buchatskaya, T. Cai, E. Rutherford, D. de Las Casas, L. A. Hendricks, J. Welbl, A. Clark, T. Hennigan, E. Noland, K. Millican, G. van den Driessche, B. Damoc, A. Guy, S. Osindero, K. Simonyan, E. Elsen, O. Vinyals, J. Rae, and L. Sifre, "An empirical analysis of compute-optimal large language model training," in *Advances in Neural Information Processing Systems*, vol. 35, 2022, pp. 30 016–30 030.